2026 Spring – UAV Control Special Topics : Homework 1

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1.UAV Modeling

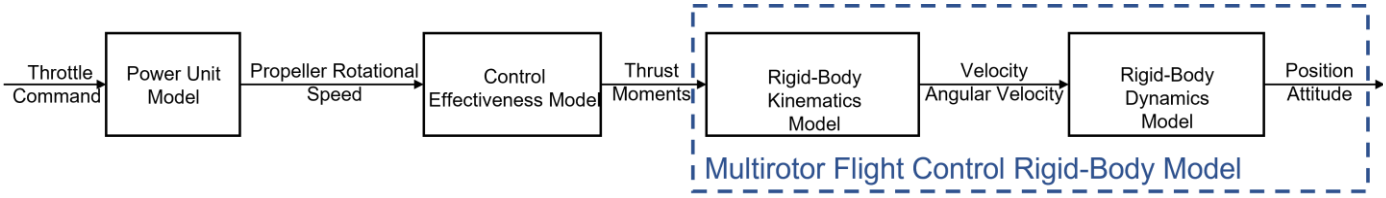


Fig1. Hierarchical Modeling Structure of a Multirotor System

In control system design, it is essential to first understand the system before developing a controller. In this project, the system refers to the UAV, while the controller represents only one component of the overall architecture.

Figure 1 illustrates the fundamental multirotor model, describing the process from throttle input to changes in the UAV's attitude and position. Among these components, the multirotor rigid-body flight dynamics model plays a central role.

Depending on the chosen coordinate transformation convention, three different model representations can be derived.

Euler Angle:

$$
\begin{aligned}
{}^{e}\dot{p} &= {}^{e}v = R \cdot {}^{b}v \\
{}^{e}\dot{v} &= -[{}^{b}\omega]_{\times} \cdot {}^{b}v + fRe_3/m \\
\dot{\boldsymbol{\theta}} &= W \cdot {}^{b}\omega \\
J \cdot {}^{b}\dot{\omega} &= -{}^{b}\omega \times (J \cdot {}^{b}\omega) + G_a + \tau
\end{aligned}
$$

Rotational Matrix:

$$
\begin{aligned}
{}^{e}\dot{p} &= {}^{e}v = R \cdot {}^{b}v \\
{}^{e}\dot{v} &= -[{}^{b}\omega]_{\times} \cdot {}^{b}v + fRe_3/m \\
\dot{R} &= R[{}^{b}\omega]_{\times} \\
J \cdot {}^{b}\dot{\omega} &= -{}^{b}\omega \times (J \cdot {}^{b}\omega) + G_a + \tau
\end{aligned}
$$

Quaternion:

$$
\begin{aligned}
{}^{e}\dot{p} &= {}^{e}v = R \cdot {}^{b}v \\
{}^{e}\dot{v} &= -[{}^{b}\omega]_{\times} \cdot {}^{b}v + fRe_3/m \\
\dot{q}_0 &= -\frac{1}{2} q_v^T \cdot {}^{b}\omega \\
\dot{q}_v &= \frac{1}{2}(q_0 I_3 + [q_v]_{\times})^{b}\omega \\
J \cdot {}^{b}\dot{\omega} &= -{}^{b}\omega \times (J \cdot {}^{b}\omega) + G_a + \tau
\end{aligned}
$$

Notation Definition

The superscripts e and b denote quantities expressed in the earth (inertial) frame and body-fixed frame, respectively.

States and Kinematics

- ${}^{e}p \in R^3$: *UAV position vector in the earth frame*
- ${}^{e}v \in R^3$: *UAV linear velocity in the earth frame*
- ${}^{b}v \in R^3$: *UAV linear velocity in the body frame*
- ${}^{b}\omega \in R^3$: *UAV angular velocity expressed in the body frame*
- $R \in SO(3)$: *rotation matrix from body frame to earth frame*
- $W(\theta) \in SO(3)$: *transformation matrix*

  mapping body angular velocity to Euler angle rates
- $\boldsymbol{\theta} = [\phi, \theta, \psi]^T$: Euler angles (roll, pitch, yaw)

Dynamics and Parameters

- m: *mass of the UAV*
- $J \in R^{3\times3}$: inertia matrix expressed in the body frame
- f: total thrust generated by propellers (along body $z-$axis)
- $\tau \in R^3$: *control torque vector*
- G_a: *Gyroscopic Torque*
- $e_3 = [0,0,1]^T$: *unit vector along the $z-$axis*
- $[\cdot]_\times$: skew $-$ symmetric matrix operator such that $[a]_\times b = a \times b$

For Quaternion Representation

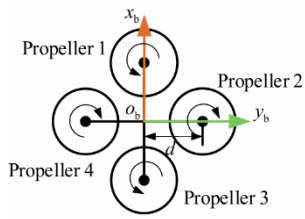
- $q = [q_0, q_v]^T$: *unit quaternion*, q_0 : *scalar part* , $q_v \in R^3$ *vector part*
- I_3 : 3×3 identity matrix

For simplicity, the following assumptions are made in modeling the multirotor:

- The multirotor is treated as a rigid body.
- The mass and the inertia tensor are constant.
- The geometric center of the multirotor coincides with its center of mass.
- The multirotor is subjected only to gravity and propeller thrust.

  Gravity acts along the positive direction of the $o_e z_e$ axis,

  while the propeller thrust acts along the negative direction of the $o_b z_b$ axis.



- Propellers with odd indices rotate counterclockwise, while propellers with even indices rotate clockwise.

Compared with other rigid-body dynamic models, the main distinction of this model is that the thrust generated by the propellers is always perpendicular to the vehicle plane.